INCH-POUND
MIL-DTL-24643/18G
27 November 2023
SUPERSEDING
MIL-DTL-24643/18F
1 October 2009

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +221 DEGREES FAHRENHEIT (+105 DEGREES CELSIUS), 1,000 VOLTS, TYPE LSMSCU (INCLUDING VARIATION LSMSCS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

This cable can be manufactured in two variations: LSMSCU and LSMSCS (double overall shield).

Construction (Watertight with Circuit Integrity)

First - Copper conductor, coated or uncoated, 18 AWG, class B stranding.

Second - Thermoset insulation. Minimum average wall thickness of 0.018 inch. Minimum diameter of the insulation 0.084 inch.

Third - An optional glass braid.

Fourth - An optional covering (unless required by 3.3.6 of MIL-DTL-24643). Diameter over the glass braid covering shall be a maximum of 0.113 inch. Standard identification code applied by method 1. Outer surface to be printed shall appear black.

Fifth - The required number of conductors (see table I or table II) shall be cabled together in accordance with 3.4.4 of MIL-DTL-24643, except for the lay of the two outer layers for sizes 44, 61, and 91, which shall not exceed 18 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward. Fillers may be used as necessary to form a firm, well-rounded assembly.

Sixth - An optional binder.

Construction for Variant Type LSMSCU

Seventh - Cross-linked polyolefin jacket. See table I or table II for thickness.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

Construction for Variant Type LSMSCS

Seventh - Braided shield of tin-coated copper.

Eighth - Spirally wrapped 0.002-inch-thick polyethylene terephthalate (polyester film) separator tape with 25 percent minimum overlap.

Distribution Statement A. Approved for public release: distribution is unlimited.

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Ninth - Braided shield of tin-coated copper.

Tenth - An optional binder.

Eleventh - Cross-linked polyolefin jacket. See table I or table II for thickness.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

TABLE I. Details (unshielded).

Military part no. M24643/18	Type and size	No. of conductors	Jacket thickness min. avg. (inch)	Overall diameter (inches)	
				min.	max.
-01UN	LSMSCU-7	7	0.040	0.447	0.484
-02UN	LSMSCU-10	10	0.050	0.575	0.622
-03UN	LSMSCU-14	14	0.050	0.617	0.668
-04UN	LSMSCU-19	19	0.050	0.682	0.738
-05UN	LSMSCU-24	24	0.050	0.790	0.855
-06UN	LSMSCU-30	30	0.050	0.833	0.901
-07UN	LSMSCU-37	37	0.060	0.926	1.002
-08UN	LSMSCU-44	44	0.060	1.030	1.114
-09UN	LSMSCU-61	61	0.060	1.156	1.250
-10UN	LSMSCU-91	91	0.060	1.369	1.480

TABLE II. Details (shielded).

Military part no. M24643/18	Type and size	No. of conductors	Jacket thickness min. avg. (inch)	Overall diameter (inches)	
				min.	max.
-01UD	LSMSCS-7	7	0.040	0.507	0.544
-02UD	LSMSCS-10	10	0.050	0.635	0.682
-03UD	LSMSCS-14	14	0.050	0.677	0.728
-04UD	LSMSCS-19	19	0.050	0.742	0.798
-05UD	LSMSCS-24	24	0.050	0.850	0.915
-06UD	LSMSCS-30	30	0.050	0.893	0.961
-07UD	LSMSCS-37	37	0.060	0.986	1.065
-08UD	LSMSCS-44	44	0.060	1.090	1.174
-09UD	LSMSCS-61	61	0.060	1.216	1.310
-10UD	LSMSCS-91	91	0.060	1.429	1.540

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REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	7.07
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	3,000
Conductor to shield (type LSMSCS only)	1,500
Overall shield to overall shield (type LSMSCS only)	200
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to conductor	500
Conductor to shield (type LSMSCS only)	500
Overall shield to overall shield (type LSMSCS only)	100
Conductor continuity	No failure
Shield continuity (type LSMSCS only)	No failure

Group A:

Visual and dimensional	No failure
Watertightness (see MIL-DTL-24643 for limits of water leakage)	No failure
Abrasion resistance (insulation) (scrapes, min.)	250
Crack resistance (insulation)	No damage

Group B:

Cold bending (cable)	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 392 °F [200 °C])	50
Drip (203±1.8 °F [95±1 °C])	No failure
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150

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Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35

Group C:

Physicals (aged) (air oven)	
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (insulation) (cycles, min.)	25
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Cable fill (sealant) removability	No failure
Shrinkage	No failure
Surface transfer impedance (type LSMSCS only)	
Milliohms per meter (max.)	700
Electromagnetic pulse (EMP) response (dB, min.)	60

Group D:

Gas flame (circuit integrity) (1 hour)	No failure

NOTE: Type LSMSCS only, shields not grounded during gas flame test.

Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Cold working (-4±3.6 °F [-20±2 °C])	No damage
Gas flame (circuit integrity) (3 hours)	No failure

NOTE: Type LSMSCS only, shields not grounded during test.

Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18

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Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSMSCU, LSMSCS (sizes 7 and 10)	1,500
LSMSCU, LSMSCS (all other sizes)	1,000

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-034)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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